Colt Melton

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EDUCATION

Kettering University | Flint, MI Expected January 2027
Bachelor of Science, Computer Science

Ajou University | Suwon, South Korea February 2025 – July 2025
Exchange Program, Software Engineering

TECHNICAL SKILLS

Languages: Python, C#, C++, SQL, JavaScript
Systems and Data: CAN bus, OPC UA, REST APIs, WebSockets, Kafka, PostgreSQL, Linux
ML and Engineering: PyTorch, SciPy, signal processing, YOLOv8, Docker Compose, VMware, Bamboo, Robot Framework, Prometheus, Grafana, Git
Enterprise Software: SAP, ERP audit support, Linux scripting, SQL queries, data aggregation

PROFESSIONAL EXPERIENCE

CSI Intern October 6, 2026 – Present
Central Garden & Pet
• Support audit-related corporate software work in SAP and enterprise resource planning systems.
• Write scripts and automate Linux workflows to support corporate software operations and audit tasks.
• Develop SQL queries and data aggregation software to organize and summarize information for audit-related work.

Research & Development Engineer / Scientist Intern October 2025 – October 6, 2026
Endress+Hauser | Ann Arbor, MI
• **Developed a spectroscopy monitoring platform** combining OPC UA telemetry, live spectral visualization, and analyzer diagnostics across **30+ industrial analyzers**.
• **Automated deployment and validation workflows** with Bamboo, VMware, and Robot Framework, eliminating approximately **4 hours of manual work per release cycle** across multiple analyzer configurations.
• **Built diagnostic and analysis tooling** with lead spectroscopy scientists for Raman analyzer validation, analytical-method development, and customer-support investigations.

Automation Software and Calibration Engineer Intern October 2023 – July 2025
Bosch International Engineering | Farmington Hills, MI
• **Built automation software** with proprietary engineering APIs to analyze ASCET-generated control software, enabling engineers to trace parameter dependencies and inspect generated C-code relationships.
• **Investigated prototype and production braking-system behavior** through CAN telemetry, diagnostic tooling, and vehicle testing, tracing interactions among calibration parameters, embedded control logic, ECU interventions, and vehicle dynamics.
• **Supported safety-critical prototype integration** through CAN diagnostics, instrumentation, wiring-harness fabrication, connector termination, and electrical troubleshooting.

IT Security & Analytics Work-Study July 2025 – Present
Kettering University | Flint, MI
• **Analyzed simulated-phishing campaign data** to identify user-risk trends and security-awareness gaps, then translated click, report, and repeat-risk rates into department-level training priorities.
• **Built executive reporting** that summarized campaign performance, highlighted recurring weaknesses, and converted security-event data into actionable recommendations.

RESEARCH

Machine-Learning-Guided Raman Spectral Decomposition Senior Thesis
• **Designed and evaluated a 2.9M-parameter PyTorch 1D U-Net** that decomposes Raman spectra into baseline and overlapping peak components across 512-point windows using a **sample-grouped 70/20/10 train/validation/test split**.
• **Built an end-to-end analysis pipeline** for peak detection, physics-aware fitting, matching, residual analysis, and downstream calibration across **602 spectra and 19,753 pure-peak matches**.
• **Measured reconstruction R^2 above 0.95** and ethane/propane calibration R^2 up to 0.98; retained residual and overlap analyses to expose failure behavior.

SELECTED PROJECTS

Distributed Telemetry Pipeline github.com/coltmelton/Metringest
• **Built a Docker Compose telemetry system** with FastAPI ingestion, Kafka buffering, asynchronous validation, PostgreSQL persistence, Prometheus/Grafana observability, and a React operations dashboard.
• **Injected duplicate, late, malformed, and failed device events** and routed invalid workloads through dead-letter handling to exercise debugging and recovery paths.

UAV Object Detection and Tracking github.com/coltmelton/UAV-Object-Detection-Tracking
• **Built an experimental aerial-video pipeline** that accepts webcam, IP-stream, and DJI-oriented inputs, performs YOLOv8 detection and multi-object tracking, and emits approximate local coordinates.
• **Implemented tunable WebSocket streaming** with frame-rate, resolution, and JPEG-quality controls; evaluation, georeferencing, and track-loss characterization remain active work.